

```

1  --s15_3 behavioral implementation
2
3  LIBRARY ieee;
4  USE ieee.std_logic_1164.ALL;
5  USE ieee.numeric_std.ALL;
6
7  ENTITY s15_b IS
8      PORT (
9          clk : IN STD_LOGIC;
10         x : OUT STD_LOGIC_VECTOR(1 DOWNTO 0)
11     );
12 END ENTITY;
13
14 ARCHITECTURE behavioral OF s15_b IS
15     SIGNAL cnt : UNSIGNED(1 DOWNTO 0) := (OTHERS => '0');
16     SIGNAL count_up : STD_LOGIC := '0';
17 BEGIN
18
19     --flipflops
20     ff : PROCESS (clk)
21     BEGIN
22         IF rising_edge(clk) THEN
23             IF count_up = '1' THEN
24                 IF cnt = "11" THEN
25                     count_up <= '0';
26                 ELSE
27                     cnt <= cnt + 1;
28                 END IF;
29             ELSE
30                 IF cnt = "00" THEN
31                     count_up <= '1';
32                 ELSE
33                     cnt <= cnt - 1;
34                 END IF;
35             END IF;
36         END IF;
37     END PROCESS;
38
39     --output logic
40     x <= STD_LOGIC_VECTOR(cnt);
41 END ARCHITECTURE behavioral;

```